

Summary

ML Research Engineer focused on post-training and generative models: RL fine-tuning for multimodal LLMs, VLA fine-tuning for manipulation, and generative trajectory models for autonomous driving. Experience building training and evaluation pipelines and optimizing production inference. Published at ICML, BMVC, and RecSys.

Work Experience

Scale AI

May 2025 - Present

ML RESEARCH ENGINEER

New York City, NY

- Fine-tuned π **vision-language-action models (openpi)** on open-source (Bridge) and Scale-collected manipulation datasets to measure data quality: tuned the **flow-matching action expert** and built visualizations to inspect the action-denoising process.
- **Post-trained multimodal Qwen3.5-9B** into a tool-using deep research agent, gaining **+12 to +19 points across public benchmarks** over base. The trained 9B matches the 27B base model despite being $3\times$ smaller.
- Built the **end-to-end RL pipeline**: GRPO/GSPO on SkyRL with a Megatron backend, verifiable rewards, and 64k-context **multimodal rollouts over text and web-page images**. Cold-started it on SFT data distilled from Qwen3.5-27B teacher rollouts.
- **Debugged distributed RL runs** that collapsed after ~ 20 steps: found importance-sampling ratios exploding to 903x, caused by a training vs. inference mismatch under sequence parallelism, then drove the **FSDP2 to Megatron** migration that fixed it.

Robot Learning in Simulation

2026

PERSONAL PROJECT

- **Trained and evaluated PPO manipulation policies** in Robosuite/MuJoCo for cube lifting; **diagnosed reward-hacking failure modes** where policies optimized shaped rewards without completing the task.

Cruise

April 2023 - May 2025

ML/ROBOTICS ENGINEER

San Francisco, CA

- **Pretrained, fine-tuned, and deployed large generative sequence models** for autonomous driving: autoregressive **trajectory generation** optimized with distillation and quantization, reducing inference latency by 66%.
- Encoded **multimodal scene embeddings** from perception and map inputs to condition the trajectory generation model; built **Transformer-based prediction models** over vectorized graph-structured scene representations, improving train/inference throughput by 33%.
- Designed **imitation-learning models**, RL rewards, and auxiliary losses for trajectory scoring and generation, reducing possible collisions by 13% and surfacing 90%+ of pedestrian edge cases.

Computer Vision Research Assistant

April 2020 - Nov 2020

BAIR LABS, UC BERKELEY

Remote

- Achieved **higher diversity in Generative models** by alleviating Mode Collapse in GANs caused by Catastrophic Forgetting, with an *increase in Inception Score by 0.3 and decrease in FID by 2.8*. **Accepted at BMVC 2021**.

Publications & Writing

- Coverage, Not Averages: Semantic Stratification for Trustworthy Retrieval Evaluation | *ICML 2026* | [link](#)
- Can Linguistically Related Languages Guide LLM Translation in Low-Resource Settings? | *LoResMT @ EACL 2026* | [link](#)
- Reward Hacking in RL Fine-Tuning of Foundation Models | *Blog, 2026* | [link](#)
- Overcoming Mode Collapse in Generative AI by Adaptive Multi-Adversarial Training | *BMVC 2021* (co-first author) | [link](#)
- A Synthetic Data Petri Dish for Studying Mode Collapse in GANs | *BayLearn 2020* (co-first author) | [link](#)
- Goal-driven Command Recommendations for Analysts | *Oral, ACM RecSys 2020* (first author) | [link](#)

Education

University of California San Diego

Sept 2021 - March 2023

MASTER'S IN COMPUTER SCIENCE | GPA : 3.9/4.0

San Diego, CA, USA

Indian Institute of Technology Kanpur

Aug. 2016 - May 2020

BACHELOR'S IN ELECTRICAL ENGINEERING, MINOR : MACHINE LEARNING | GPA : 9.0/10.0

Kanpur, India

Skills

Languages Python • C++ • CUDA • Shell

ML / RL PyTorch • JAX • SkyRL • Megatron • FSDP2 • DeepSpeed • vLLM

RL & Policy Learning PPO • GRPO/GSPO • imitation learning • VLA fine-tuning • flow matching • reward design

Robotics / Sim MuJoCo • Robosuite • ROS • OpenCV • LiDAR / point clouds